

Literature answer: Webster operator compact maps are not cb-norm closed

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Source Question

The source paper is Chavez-Dominguez–Dimant–Galicier, *Operator p -compact mappings*, arXiv:1809.07741 [1]. In its final section, the authors recall Webster’s problem asking whether the space of operator compact maps is closed in the completely bounded norm topology. Equivalently, using the notation of the source paper, they ask whether the operator compact norm κ_∞^o and the completely bounded norm are equivalent on $\mathcal{K}_\infty^o$.

Supporting Answer

The same authors later returned to the subject in *Revisiting Operator p -Compact Mappings*, arXiv:2405.18571 [2]. In the introduction they explicitly say that their comparison of compactness notions answers the Webster question emphasized in [1]. The decisive passage is in the discussion after their two-sided multiplication example: for $p = \infty$, the multiplication map

$$M(a, b) : \mathcal{S}_\infty \longrightarrow \mathcal{S}_\infty$$

constructed there is not operator compact, but it is the completely bounded norm limit of finite-rank truncations, and finite-rank maps are operator compact.

Conclusion

Thus Webster’s cb-norm-closedness question from arXiv:1809.07741 has a negative answer. The class of operator compact maps need not be closed in the cb-norm topology, and consequently κ_∞^o and $\|\cdot\|_{cb}$ are not equivalent on $\mathcal{K}_\infty^o$ in general.

This packet only records the literature answer to that final Webster problem. The separate tensor-norm association questions involving $/d_p^o$ and surjective hulls are not settled by this identification.

References

- [1] J. A. Chavez-Dominguez, V. Dimant, and D. Galicer, *Operator p -compact mappings*, arXiv:1809.07741, 2018.
- [2] J. A. Chavez-Dominguez, V. Dimant, and D. Galicer, *Revisiting Operator p -Compact Mappings*, arXiv:2405.18571, 2024.